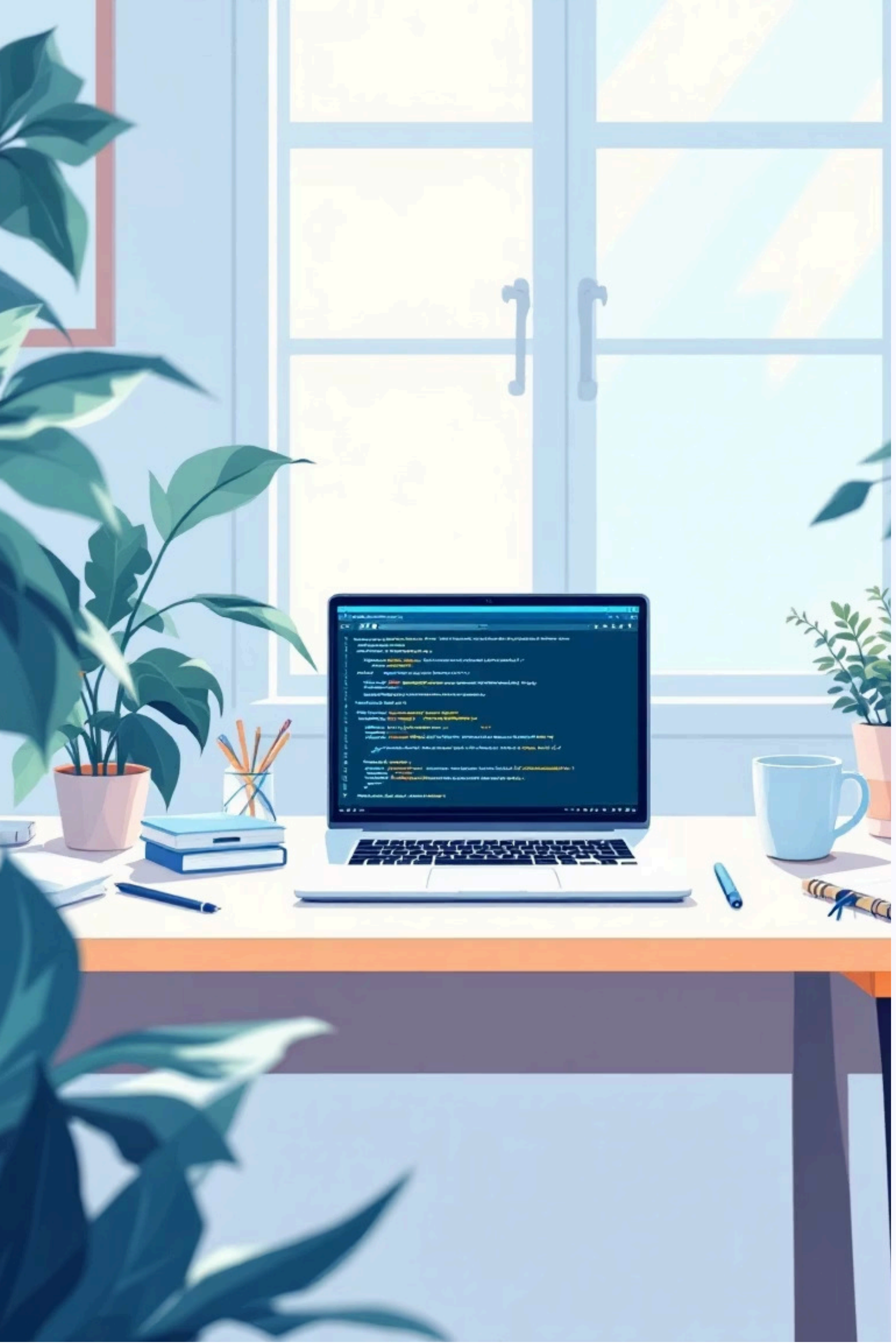


My Project - Development Tasks and Learning

In this project I worked on different development tasks, each one with a different goal. The idea was to practice real tools and understand how they work in practice.

What I had to do

I worked on many different tasks:

FastAPI - HTTP server

I built a small FastAPI server with two endpoints: one saves JSON data into a local file, and the second returns the last 10 saved records. I tested it using the browser and terminal.

Typer - CLI tool

I created a command-line app with two commands: one command adds JSON to a file, and another prints the last 10 items. I practiced running it directly from the terminal.

VitePress - documentation site

I created a simple documentation website with two pages: a landing page for a fake tool, and an installation page. I learned how static docs work.

Dockerfile + ffmpeg

I wrote a Dockerfile, installed ffmpeg inside the image, and added an endpoint/script that runs ffmpeg commands inside the container.

More Tasks I Completed

Python + Docker SDK

I used Python to run a BusyBox container, kept it alive with sleep, and then executed a command inside it to get the container hostname.

Coordinate conversion

I wrote a function that converts coordinates from Decimal Degrees (DD) into Degrees-Minutes-Seconds (DMS) based on the examples.

Markdown → PDF

I created code that converts a Markdown file into a PDF. After issues with WeasyPrint, I switched to ReportLab which worked better.

NumPy histogram

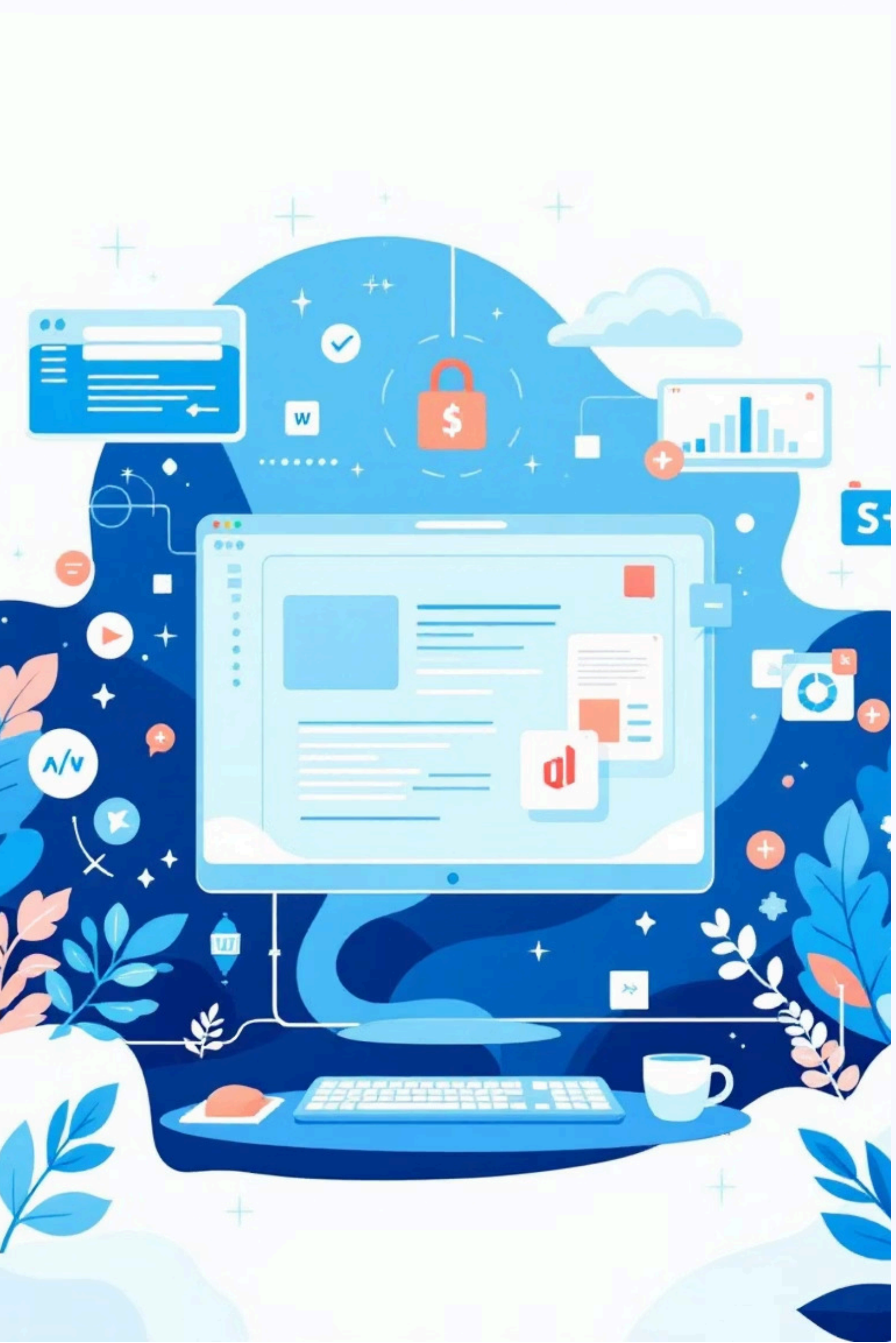
I loaded an image, read the RGB values, and built color histograms with NumPy.

Password strength checker

I wrote a function that checks a password against several rules (length, digits, uppercase, lowercase, special chars, repetitions and sequences).

K3s Markdown

I wrote a short document explaining what K3s is, when and why to use it, and how it relates to Kubernetes.



My work environment

Technologies I used

- FastAPI (server)
- Typer (CLI)
- Docker + BusyBox
- ffmpeg
- NumPy
- ReportLab
- Markdown / VitePress

I worked mainly with:

- VS Code
- Python
- Git and GitHub
- Docker
- PowerShell / Terminal
- Markdown and VitePress

My way of working

Working step-by-step helped a lot.

1. Read the requirements
2. Start with a basic version
3. Test and fix errors
4. Improve the solution
5. Commit and push to GitHub

Things I learned

how to build a simple server

how to use Git and GitHub correctly

how CLI tools work

how to work with images and files

how to convert and generate documents

how containers and Docker work

how to debug and search for solutions

how to run a BusyBox container, keep it alive using sleep, and run commands inside it

Problems I had and how I solved them

I had some issues with tools and libraries.

Problem 1

One problem was converting Markdown to PDF – I mixed two solutions and the code didn't work and it took me time to figure this out.

Problem 2

Another problem was WeasyPrint on Windows – it needed system libraries that I didn't have.

Solution: I solved it by simplifying the solution and switching to ReportLab. I learned that sometimes it's better to change the tool instead of fighting the problem.

Feedback from my friends

My friends reviewed my work. They said it looked **organized, clear and clean**. They liked that I worked step-by-step and that everything was easy to understand. They also said the final results looked good.

Conclusions and tips

(choose the ones you like):

- [ ] Work in small steps
- [ ] Test all the time
- [ ] Read tasks slowly before starting
- [ ] Don't be afraid to change your solution
- [ ] Ask for help when you're stuck

Project timeline

1

Learning – Yesterday

Learning about servers, Git, reviewing the project and understanding the tasks.

2

Work – Today

Working on the tasks, coding, building the presentation and fixing things.

3

Finish – Today

Final review, improvements and preparing the final version.